

Digital Forensics: Complete Overview

1 What is Digital Forensics?

 Digital Forensics is the **process of identifying, preserving, analyzing, and presenting** digital evidence in a way that it can be used in **investigations or court of law**. It deals with extracting evidence from computers, mobiles, servers, cloud, and other digital devices.

2 Why is Digital Forensics Done?

Companies, police, and cybercrime units perform digital forensics to:

- **Investigate Cybercrime** (hacking, phishing, ransomware).
- **Recover Deleted Data** (files, chats, emails).
- **Find Insider Threats** (employee fraud, data theft).
- **Incident Response** (analyzing malware attacks).
- **Legal Evidence** (court cases, law enforcement).

3 How is Digital Forensics Done? (Process / Lifecycle)

◆ Step 1: Identification

- Find potential sources of evidence → hard drives, mobiles, logs, emails.

◆ Step 2: Preservation

- Secure the device → no tampering.
- Use **write blockers** to prevent modifications.



◆ Step 3: Collection / Imaging

- Take a **forensic image** (bit-by-bit copy) of the drive.
- Never work on the original media.

◆ Step 4: Analysis

- Use forensic tools to analyze → recover files, logs, chats, network traffic, malware.
- ◆ **Step 5: Documentation**
- Document every step (tools used, hash values, timeline).
- ◆ **Step 6: Presentation**
- Present findings in simple, legally valid way (for court, management, or investigation report).

4 Types of Digital Forensics

1. **Computer Forensics** → Evidence from desktops, laptops, servers.
2. **Mobile Forensics** → Extracting call logs, WhatsApp, SMS, GPS.
3. **Network Forensics** → Capturing & analyzing network traffic (packets, logs).
4. **Database Forensics** → Checking SQL logs, unauthorized queries, data theft.
5. **Cloud Forensics** → Evidence from AWS, Azure, Google Cloud.
6. **Email Forensics** → Analyzing headers, SPF/DKIM/DMARC, phishing.
7. **Malware Forensics** → Reverse engineering and analyzing malware.

5 Chain of Custody (Very Important in Forensics)

👉 Definition:

The **chain of custody** is a **legal document** that records **who collected, handled, transferred, or analyzed** the evidence.

📌 Why it is important?

- To prove evidence has not been tampered with.
- To maintain credibility in court.
- To track ownership of evidence.

📋 Example entry:

- Evidence ID: 001
- Description: 500GB Seagate HDD from suspect PC
- Collected by: John Smith, 20-Aug-2025, 10:30 AM
- Transferred to: Lab Technician (signed, timestamped)

6 Tools Available for Digital Forensics

◆ Free / Open-source Tools

- **Autopsy** (GUI-based forensic suite)
- **Sleuth Kit** (command line forensic toolkit)
- **FTK Imager** (imaging, preview)
- **Volatility** (RAM analysis)
- **Wireshark** (network traffic analysis)
- **Magnet RAM Capture** (memory forensic)
- **Scalpel / Photorec** (file carving, deleted file recovery)

◆ Paid / Enterprise Tools

- **EnCase Forensic** (industry standard, court accepted)
- **X-Ways Forensics** (powerful, lightweight)
- **Magnet AXIOM** (modern evidence collection: PC, mobile, cloud)
- **Cellebrite UFED** (mobile forensics)
- **FTK (Forensic Toolkit)** (deep analysis, indexing)